

Electrical Power Engineering

Engineering Technology

May, 2026



Electrical Power Engineering (A06046)

Short Title:	Electrical Power Engineering	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electrical Engineering	
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Credits	5	Level: Intermediate

Description of Module / Aims

The aim of this module is to provide the student with an understanding of the electrical transmission and distribution circuits, equipment and related calculations. In addition, the student will be familiarised with the safety and maintenance requirements for electrical medium and high voltage electrical systems.

Programmes

	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 5 / M
	BEng (Hons) (WD_ETRIC_B)	2 / 4 / M
ELEC-0035	BEng in Electrical Engineering (WD_EELCT_D)	3 / 6 / M

Indicative Content

- Power generation network sources; traditional, wind power and solar.
- Power transmission topologies and equipment.
- Power distribution topologies and equipment.
- Centralised and distributed network management
- Protective relays, strategies and associated transducers.
- Preventative Maintenance & Electrical Safety
- Load, BIL, Power flow, Fault, and stability calculations
- Typical project: Design applications to calculate load unbalance and/or perform a power flow analyse for a network.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the operation, application, control and function of typical electrical transmission equipment and switchgear.
2. Describe the operation, application, control and function of typical electrical distribution equipment and switchgear.
3. Describe and compare transmission and distribution topologies.
4. Explain the preventative maintenance procedures and the safety requirements for typical medium voltage and high voltage equipment.
5. Perform typical distribution and transmission calculations, circuit design and simulation e.g. Load, BIL, Power flow, Fault, and stability calculations.
6. Demonstrate the professional, ethical and effective communication skills of a professional engineer.

Learning and Teaching Methods

- The course will consist of lectures, laboratories, mini project and formative assessments.

- The emphasis on course delivery will to engage the student in the learning of the material, through the integration of the course materials, with the real case studies in laboratories and mini project.
- The formative assignments will be used to provide an opportunity for the student to engage with the material, before it is applied, to maximise the learning opportunities for the student.
- Self-directed learning, moodle, and recommended software.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Final Practical	10%	5,6
Continuous Assessment	20%	
<i>In-Class Assessment</i>	10%	1,2,3,4
<i>Project</i>	10%	1,2,3,5,6

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Green, J.N. and R Wilson. *_Electrical Power Generation, Transmission & Distribution_*. UK: CRC Press, 2007.

Supplementary Material

- Bird, J.O. and A.J.C. May. _Standard Handbook for Electrical Engineers_. : Elsevier, 2013.
- Warne, D.F. _Electrical Power Engineering Handbook_. 2nd ed.. Oxford: Newnes, 2005.

Requested Resources

- ENGINEERING LAB: Electrical